



Dr. Dilip Bathena
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Mobile: +91 9542702606
Date of Birth: 14/07/1983
Gender: Male

Senior Capacity & Performance Manager

—Data Science, AI Ops (Prompt Engineering), SRE, On-Prem & Multi-Cloud Architecture, Technical Project Management, Multidisciplinary Research,

Professional Summary:

PMP-trained, Agile Certified, IBM Data Science Level2 Certified, Senior Capacity & Performance Manager and Research Scientist/ SRE / Hybrid Cloud Operations Architect with 15+ years of experience specializing in capacity planning, performance management, ITIL processes, and hybrid cloud operations (AWS, AZURE, VMware). Expert in Data Science (SPSS, Python, Advanced Excel, Power BI), Generative AI, Prompt Engineering, Context Engineering, Google LLM Notebooks, Azure AI, and techniques to prevent hallucinations in LLM outputs. Delivered 30% downtime reduction and 25% improvement in resource utilization through advanced performance forecasting and predictive capacity management. Ph.D. – ICT & Socio-Economic Research Development (Data Science & Research focused).

Core Competencies

- **Capacity & Performance Management** – Planning, forecasting, optimization, SLA compliance.
- **SRE Practices & Hybrid Cloud Architecture** (AWS, Azure, GCP, VMware)
- **Data Science & AI** – SPSS, Python, Advanced Excel, Power BI, **Generative AI**, Azure AI, Prompt Engineering, Hallucination Prevention
- **Infrastructure as Code (Terraform) & Automation**
- **ITIL, Incident, Change & Problem Management**
- **Project Management (PMP-trained, Prince2, Agile/Scrum)**
- **Sample AI Capacity forecast Model-built GitHub link:-**
[AI-Capacity-forecast-model/README AI Capacity Forecast Agent.md at main · dilipbathena/AI-Capacity-forecast-model](#)

Certifications

- **PMP (Project Management Professional) – Trained**
- **Prince2 | ITIL | Microsoft Azure Architect | AWS Cloud Practitioner**
- **Microsoft Azure AI Fundamentals | IBM Data Science Certified**
- **Cognizant Prompt Engineering & Context Engineering**

Professional Experience

Cognizant – Senior Capacity & Performance Manager / SRE November 2024 – Present

- Led capacity planning and performance management for hybrid cloud operations across AWS, Open shift, Azure.
- Capacity Demand Management and Optimization.
- Applied data science (SPSS, Python, Advanced Excel, Power BI) for performance forecasting and proactive resource optimization.
- Utilized Generative AI, Prompt Engineering, Context Engineering, and Azure AI for reliable monitoring while preventing hallucinations.
- Executed Terraform automation and ITIL-based incident/change management.
- Participated in Vibe Coding Hackathon and Won Guinness Record Certificate.

IBM/Kyndryl – Capacity and Performance Management and Research (Banking, Airlines, Automotive, Retail, Research domains) 2016 – 2024

- Directed SRE Practices, capacity planning and performance management accounts using advanced research techniques and Data Science.
- Achieved 30% reduction in downtime and 25% improvement in resource utilization through predictive performance forecasting and capacity modeling (SPSS, Python, Power BI).
- Applied Generative AI and Prompt Engineering expertise to build reliable AI-assisted performance monitoring solutions with effective hallucination prevention.
- Ensured SLA compliance using ITIL frameworks.

Research & Technical Projects

- Ph.D. Researcher, Nagarjuna University (2016–2022): Conducted advanced Data Science and performance modeling using SPSS, Python for ICT impact assessment.
- **NIRDPR – ICT/GIS -Multi Disciplinary Research Scientist** (2012–2016): Applied Data Science in national & international UN technical projects (GIS Mapping, Dashboards, E-Governance) for Socio-economic Development.

MYOB, Burwood (Australia) 2011

- Technical Support & System Administration with performance troubleshooting.

Technical Skills Summary

- Capacity & Performance Management: Forecasting, Optimization, SLA Compliance

- Data Science & AI: SPSS, Python, Advanced Excel, Power BI, Azure AI, Prompt Engineering, Hallucination Prevention
- Cloud & SRE: AWS, Azure, GCP, VMware, Terraform, Hybrid Cloud Operations
- Monitoring: Splunk, Datadog, Grafana, Zabbix, vROps
- Management: ITIL, PMP, Prince2, Agile

Technical Skills.

AI, Data Science & Generative AI Skills

- Azure AI Certified | Generative AI (Claude, ChatGPT, Gemini, Copilot, Bard)
- Cognizant Certified: Prompt Engineering & Context Engineering
- Python for Data Analysis (CSV, Anaconda, Spyder) & AI Scripting
- SPSS: Regression, Correlation, T-Test, Quantitative & Qualitative Analysis, Statistical Modeling & Forecasting
- Advanced Excel: Pivot Tables, Dynamic Formulas, Trends & Forecasting
- Power BI for dashboards and business insights.

Multi Cloud Infrastructure Skills

- **AWS:** EC2, RDS, IAM, VPC, Security, Performance & Capacity Management
- **VMware:** vROps, ESXi, vCenter, VM Management, Capacity & Performance (CPU, Memory, Storage, Network)
- **Hybrid Cloud:** AWS + Azure + GCP + OpenShift, Terraform IaC
- Storage & Networking: Fibre Channel, iSCSI, NFS, TCP/IP, Zabbix Monitoring
- Devops trained and Experience in Terraform Iaac.

Capacity, Performance & SRE Skills

- Capacity Planning, Performance Forecasting, Demand management & Optimization
- Monitoring: Splunk, Datadog, Grafana, Zabbix, vROps
- ITIL: Incident, Change & Problem Management

Research, Testing & Administration

- Data Science ,Research (SPSS, Python, Excel) – PhD & National/UN Projects (ISRO, CDAC)
- ICT/IT/GIS Solution Design, System Integration, Testing, & Deployment
- Exploratory, qualitative, domain, and UAT manual testing

Technical Project Management Skills:

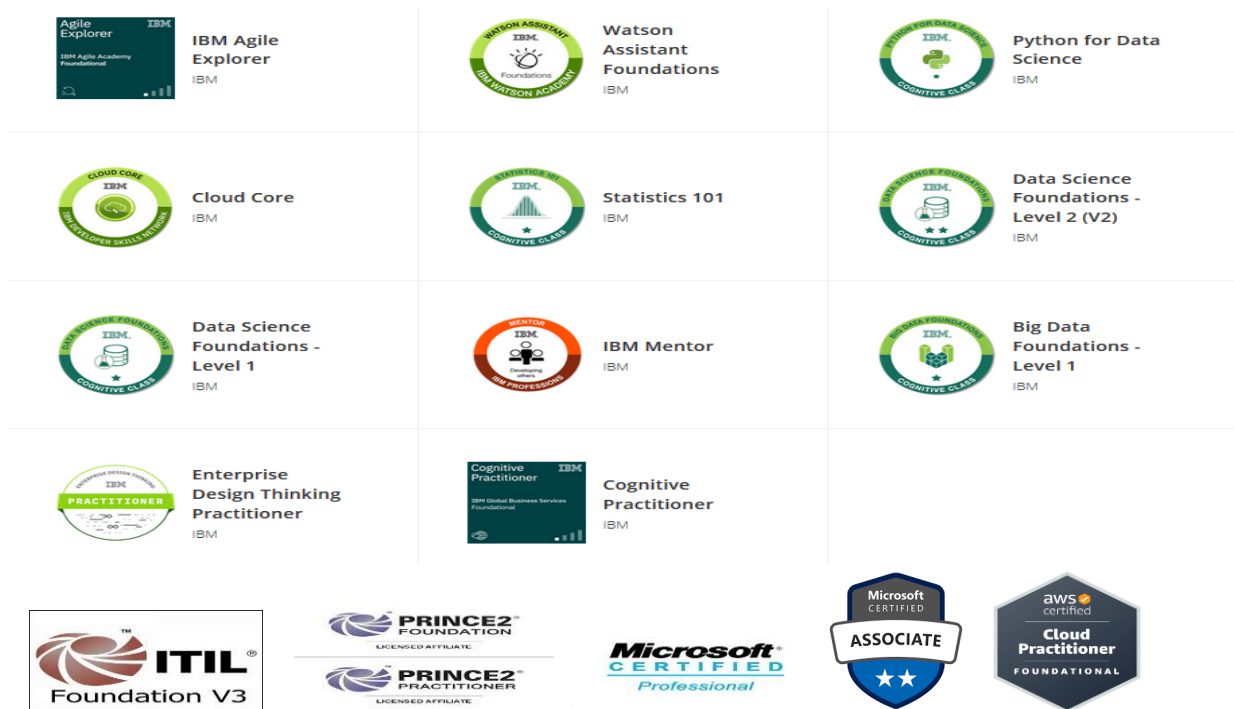
- PMP Trained, Prince2 Practioner Certified.
- Basic knowledge of SDLC, Agile, Scrum methodologies.

- ICT/IT/GIS solution design, system integration, deployment support

Systems Administration:

- Windows, Linux, Mac OS installation, configuration, deployment
- Windows Server & Linux basic administration and user management
- Technical support: remote troubleshooting (phone, email, chat, RDP)

Digital badges and Certifications



Key Achievements

- Reduced downtime by 30% and improved resource utilization by 25% through strong capacity and performance management.
- Multiple Disciplinary ICT, GIS Research publications and UN/National technical projects.

Academic Qualifications:

- Ph.D. ("Impact of Information and Communication Technology (ICT) on Socio-Economic Development of Rural Communities")
- M.Sc. (GIS), Annamalai University | M.Sc. (IT), Nagarjuna University
- B.Sc. (Computers), Nagarjuna University | M.A. (Sociology), Nagarjuna University
- PG Diploma (Rural Development), Annamalai University

- Advanced Diploma (Business Management), IT Course Holmes glen TAFE, Melbourne, Australia.
- Intermediate (MPC & Computer Science), Little Flower Jr. College (69%) | S.S.C., Little Flower High School (65%)

Professional Membership:

- Life membership of the International Sociological Association.
- Life Membership of Public relations society of India.



Projects Undertaken:

- NGO – SBI CSR Rural Health ICT Project, Coimbatore (Hospice Care) | 2016–2021
- UGC Doctoral Research Project, Nagarjuna University | 2016–2022
Impact of ICT on Socio-Economic Development of Rural Communities
- ICT Domain Projects: E-Agriculture, E-Governance, E-Banking, E-Health, E-Education
- National Institute of Rural Development and Panchayati Raj (NIRDPR) – E-Office Project | 2012–2016 (Secondary Member)
- NIRDPR – UN Research Project: CIRIDAP Rural Tech Centres Design | 2012–2016 (Primary Member)
- NIRDPR – UN Research Project: GIS-APIB Mapping & Plotting | 2012–2016 (Primary Member)
- NIRDPR – ICT/GIS Watershed Management Project | 2012–2016 (Primary Member)
- MGNREGA Social Audit & Dashboard Project | 2012–2016 (Secondary Member)
- Smart City Project (IBM) | 2016–2017 (Primary Member)

Patents / Innovations / Models / Application Development:

- Developed models in energy-efficient computing, ICT for socio-technology, and load balancing
- Conceptualized education models integrating technology and social frameworks
- Contributions to ICT-based system design, optimization, and implementation

- Multiple research papers and technical presentations in national and international forums

Presentations:

- “Green Computing – DILIP Model (Development of IT with Low Investment Procedure)”
National Conference, Vasavi College of Engineering (ISBN: 978-93-82570-58-5) | 2015
- “The Real Purpose of Education” – International Seminar, Nagarjuna University | 2015
- “Role of ICT Among Weavers During COVID-19” – UGC National Seminar | 2021

Paper Publications:

- “Green Computing – DILIP Model” – NCEITS Conference Journal (ISBN) | 2015
- “The Real Purpose of Education” – International Multidisciplinary Journal (ISSN: 2349-5774) | 2015
- “ICT for Digital Banking & E-Commerce in Rural Communities” – Indian Journal of Applied Research (ISSN: 2249-555X) | 2021
- “Perception of ICT in Rural Communities” – IJMER (ISSN: 2277-7881) | 2021
- “Willingness to Pay for E-Banking Services” – Journal of Composition Theory (ISSN: 0731-6755) | 2021
- “ICT Usage, Happiness & Stress in Agriculture Sector” – IJCRT (ISSN: 2320-2882) | 2022
- “ICT Benefits in Agriculture Across Socio-Economic Groups” – IJMER | 2022
- “ICT Usage Among Weavers” – IJMER | 2022
- “E-Banking Information Sources in Rural Communities” – IJMER | 2022
- “Population Explosion & Community Health (SDG-3)” – UGC Proceedings (ISBN) | 2023
- “Social Equality – Jyotirao Phule Model” – UGC Proceedings | 2023
- Additional publications on child marriage, social policy, and rural sanitation (UGC seminars)

Books Published:

- *Socio-Economic Dimensions and ICT* – ISBN Registered

Geographical Exposure:

- **India:** Andhra Pradesh, Assam, Chandigarh, Delhi, Haryana, Himachal Pradesh, Jharkhand, Karnataka, Kerala, Maharashtra, Odisha, Punjab, Tamil Nadu, Telangana, West Bengal
- **International:** Australia (Sydney, Melbourne, Bendigo, Ballarat), Singapore, Malaysia, Dubai